COS20028 – BIG DATA ARCHITECTURE AND APPLICATION
LAB WEEK 01

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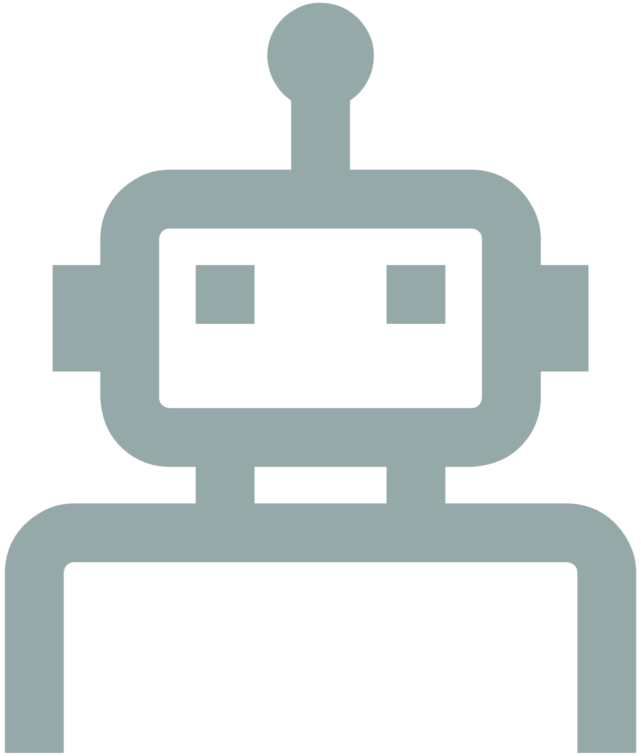
SWINBURNE UNIVERSITY OF TECHNOLOGY

Week 1

Tutorial

VM Setup

Hadoop Commands Warm Up



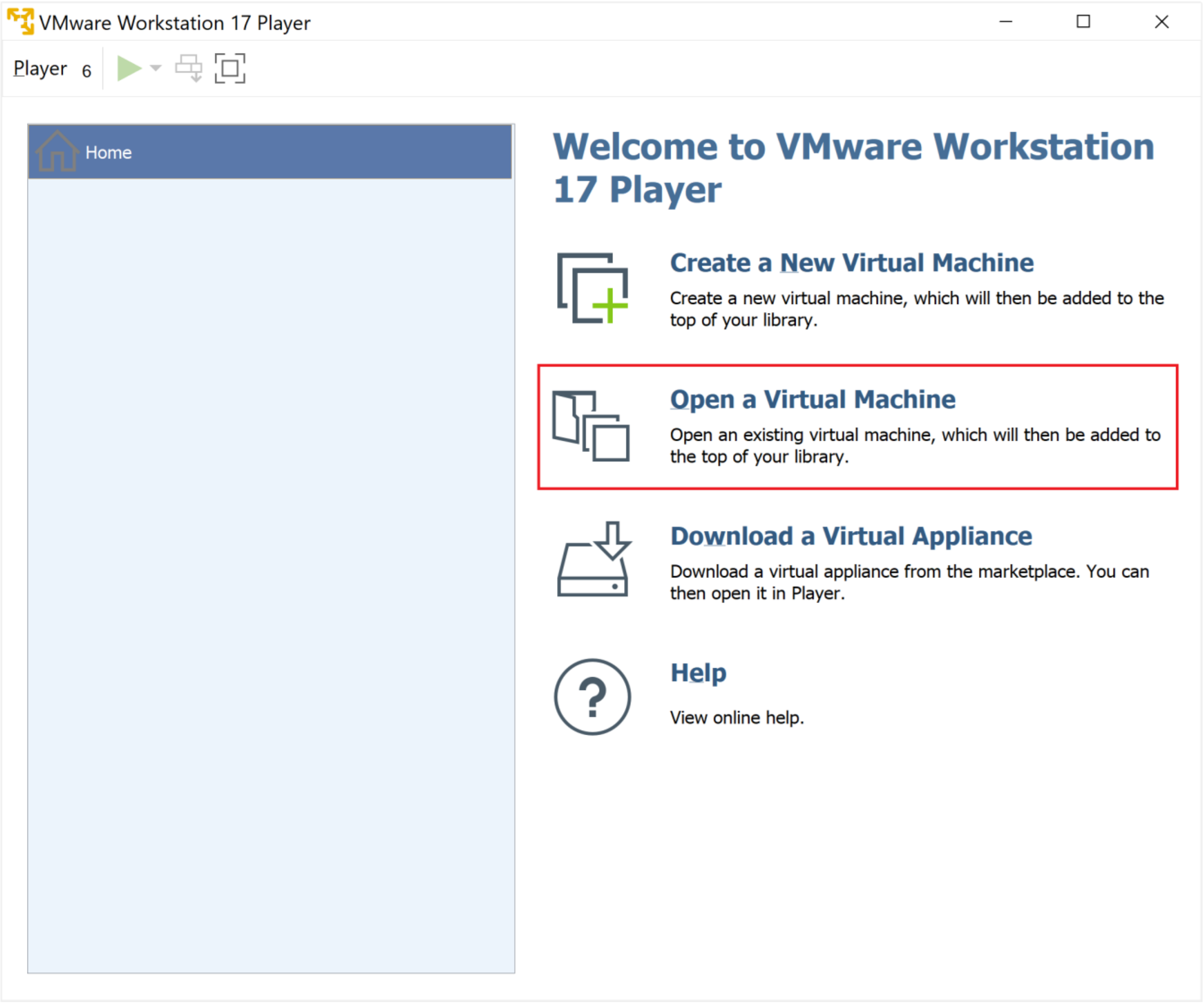
What is a Virtual Machine?

- A virtual machine is a simulated environment to be hosted by your physical machine.
- It can have different Operating Systems (OS), applications, and hardware than your physical machine.
- It is usually used as a sandbox, an experimental environment, or an agent in practice.

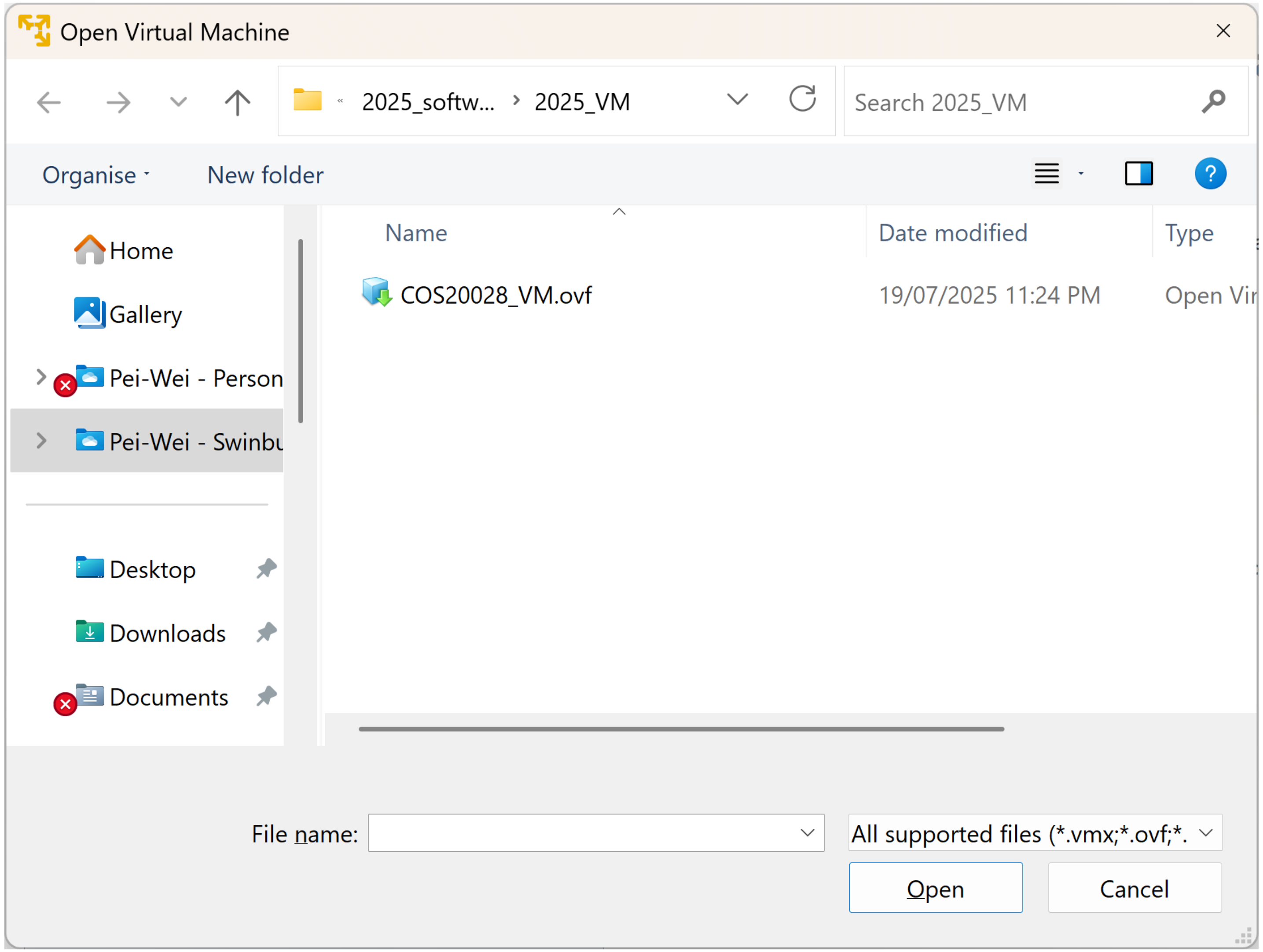
VMware Workstation Player Setup

1. Install Vmware Workstation Player or Vmware Fusion. Learn how to download following: https://knowledge.broadcom.com/external/article?articleNumber=368734
   or
2. Download the VM Image in Canvas.
   Module → COS20028 Tools → Cloudera VM

Import the VM Image

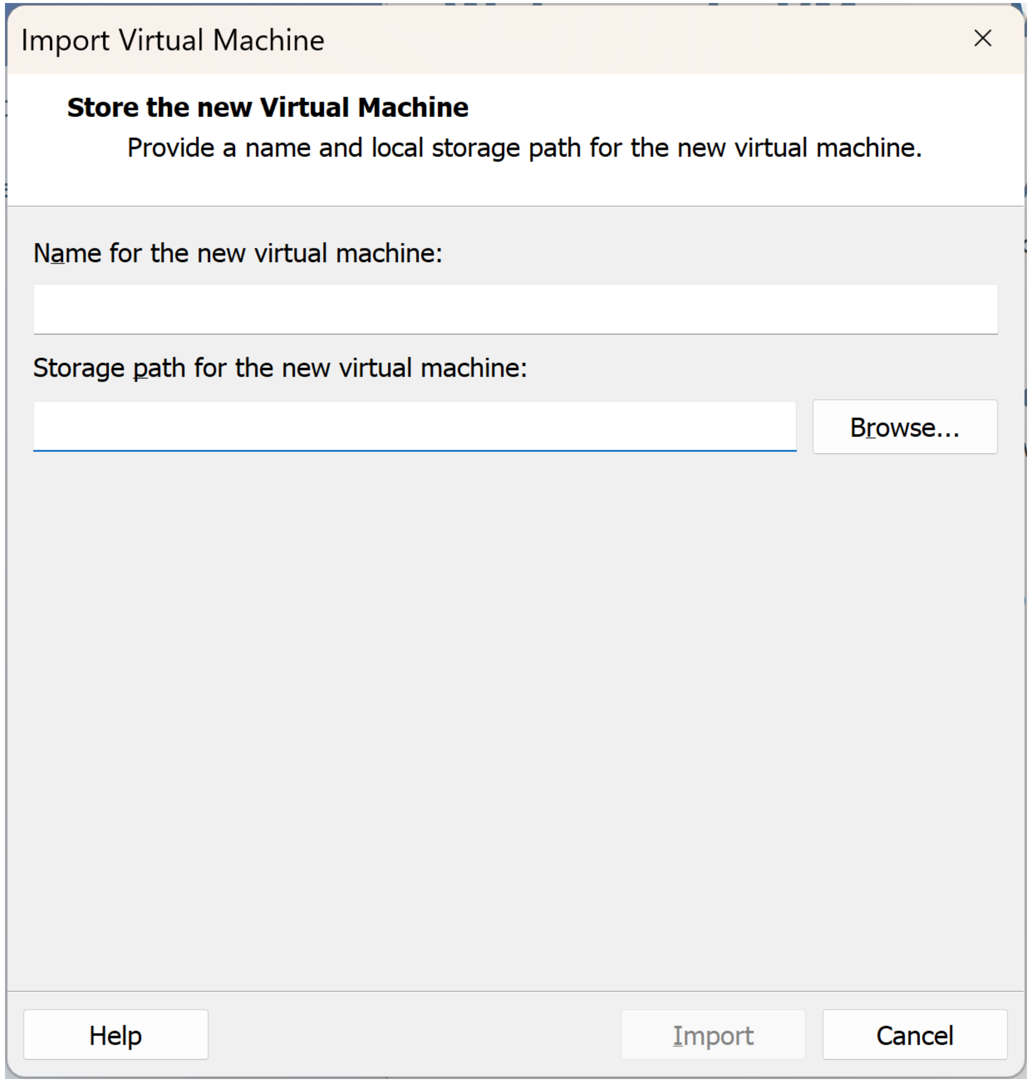


Select the Image



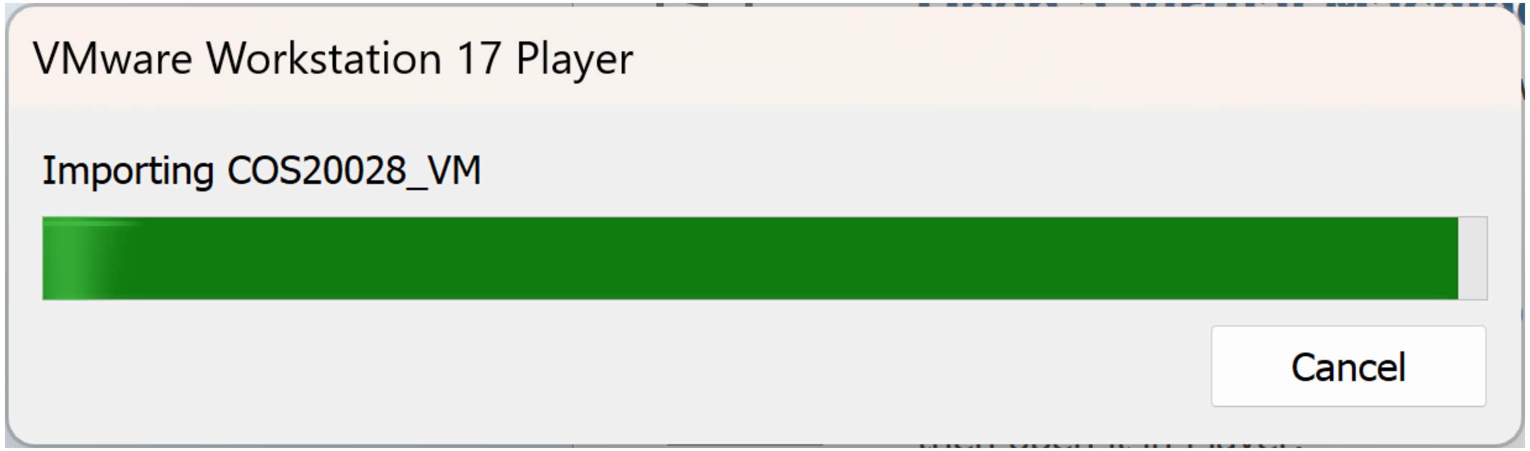
Name Your Virtual Machine

- Type in the "Name for the new virtual machine". The storage path will automatically be filled. Click "Import" to load the image.



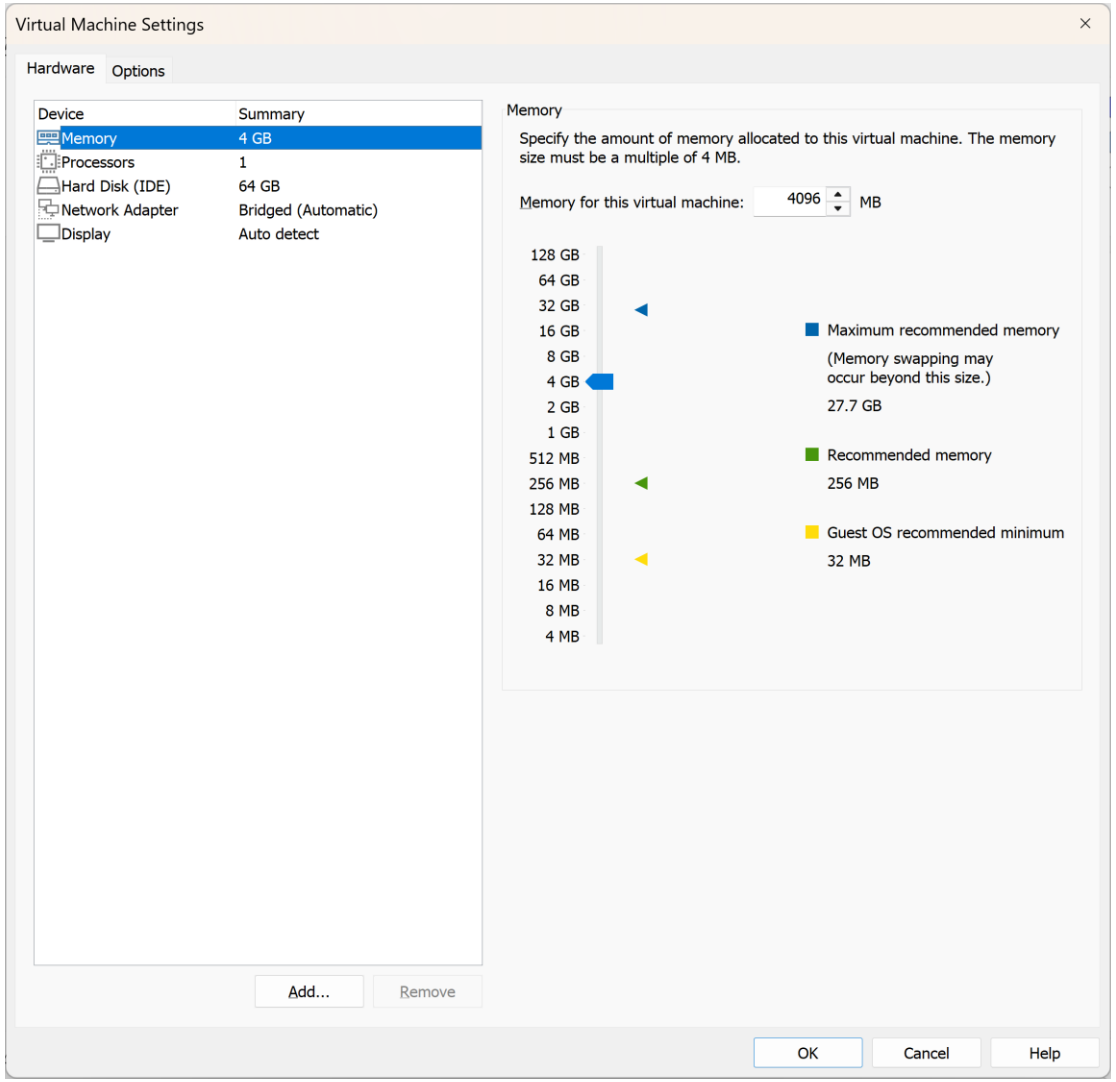
Let the Software Setup Your Machine

- Give it some time to load the selected image.



Setup Your VM, or Execute it.

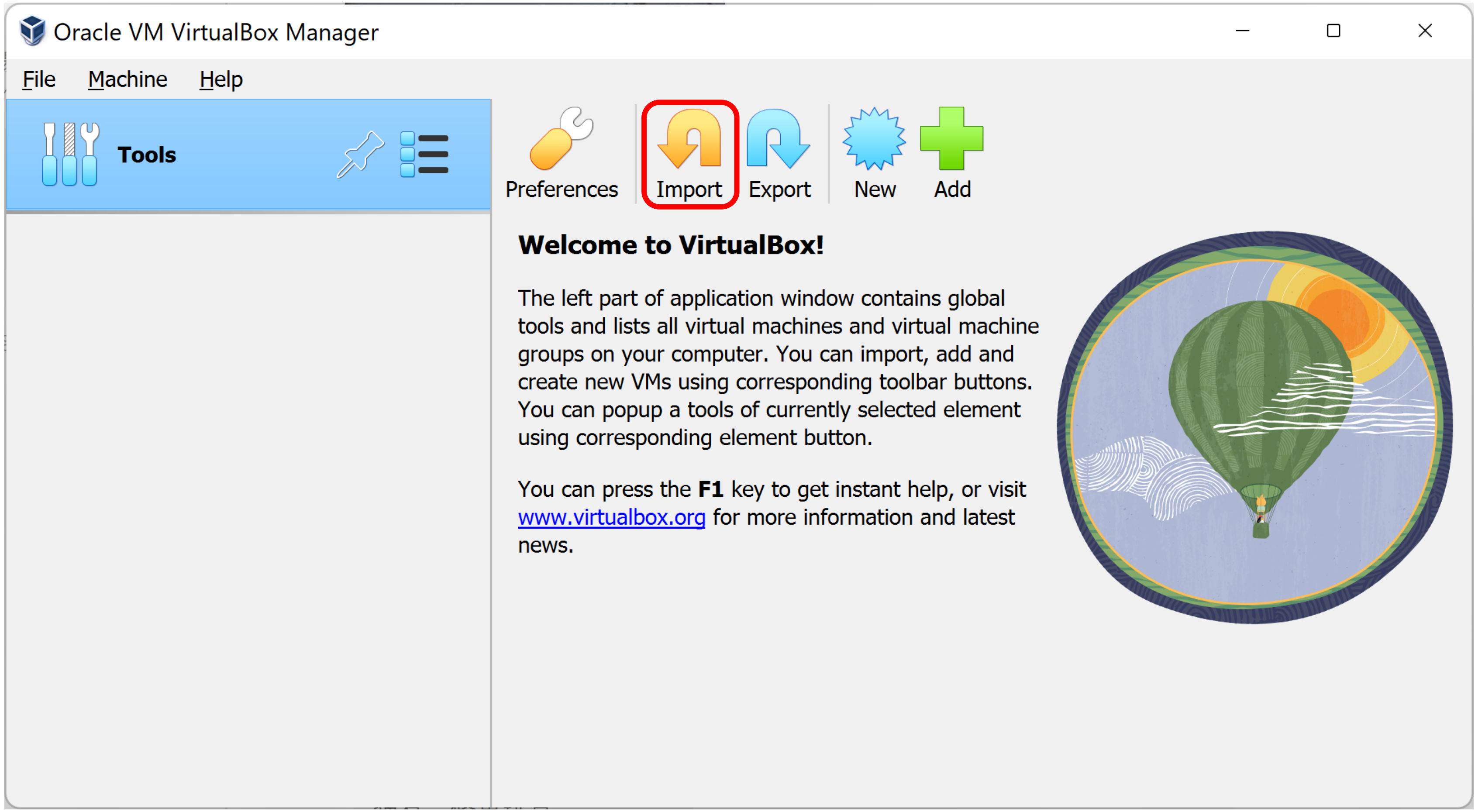
- Double-click on the machine's name to start it. You can also use "Edit virtual machine settings" to change some resource-occupying details.



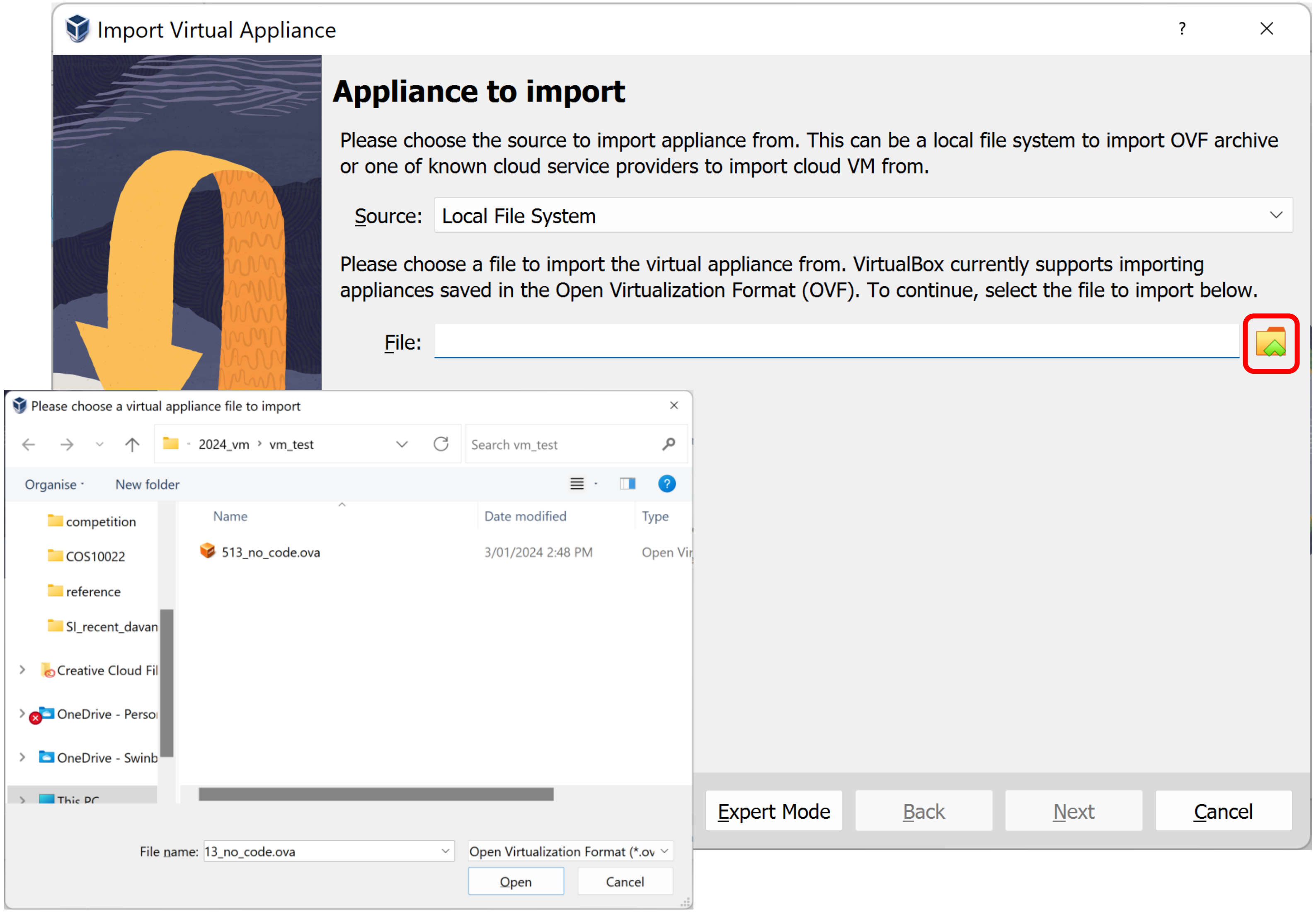
Oracle VM VirtualBox Setup

1. Install Oracle VM VirtualBox.
   Download from
   https://www.virtualbox.org/wiki/Downloads
   or
2. Download the VM Image in Canvas.
   Module → COS20028 Tools → Cloudera VM

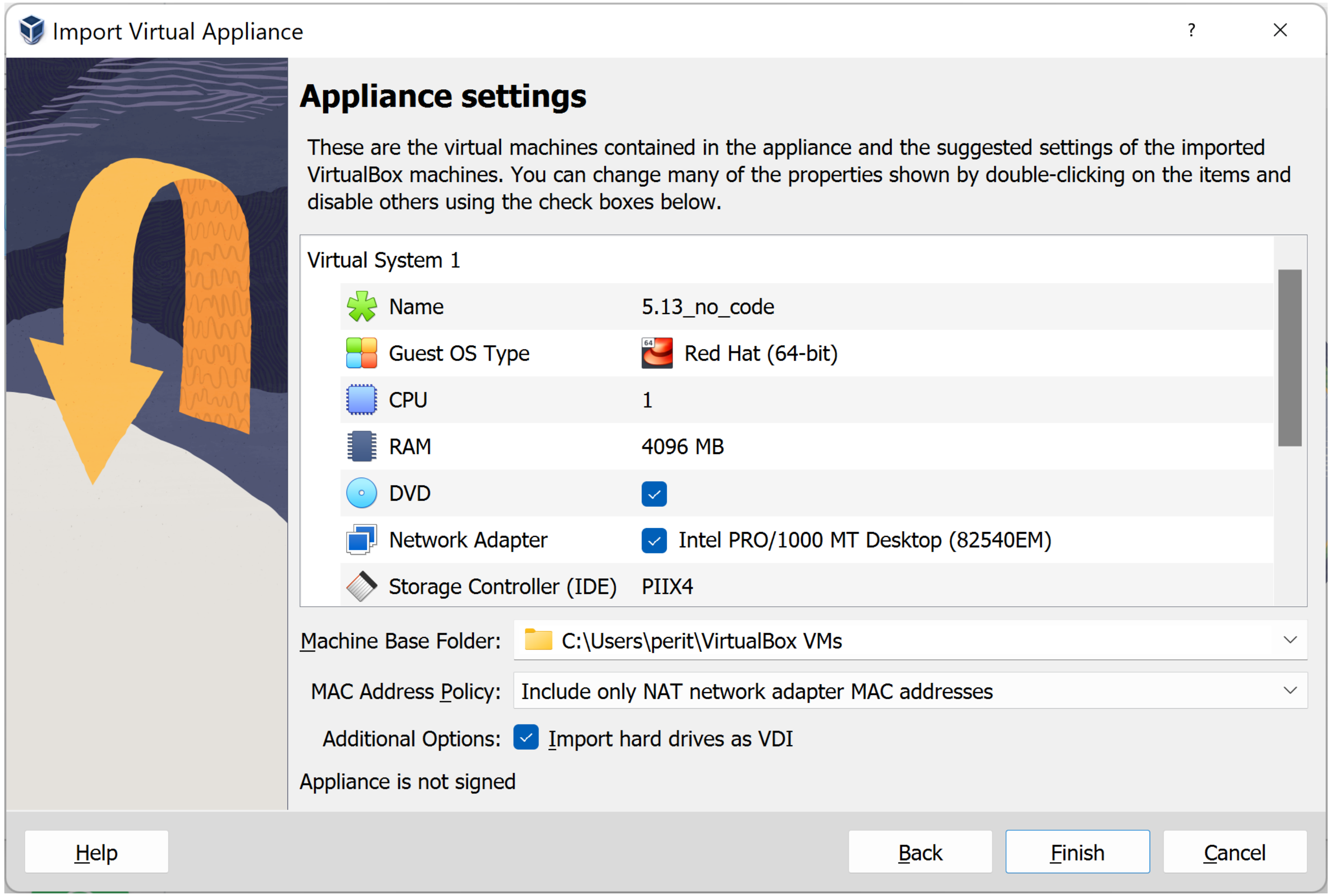
Import the VM Image



Import VM Image (2)

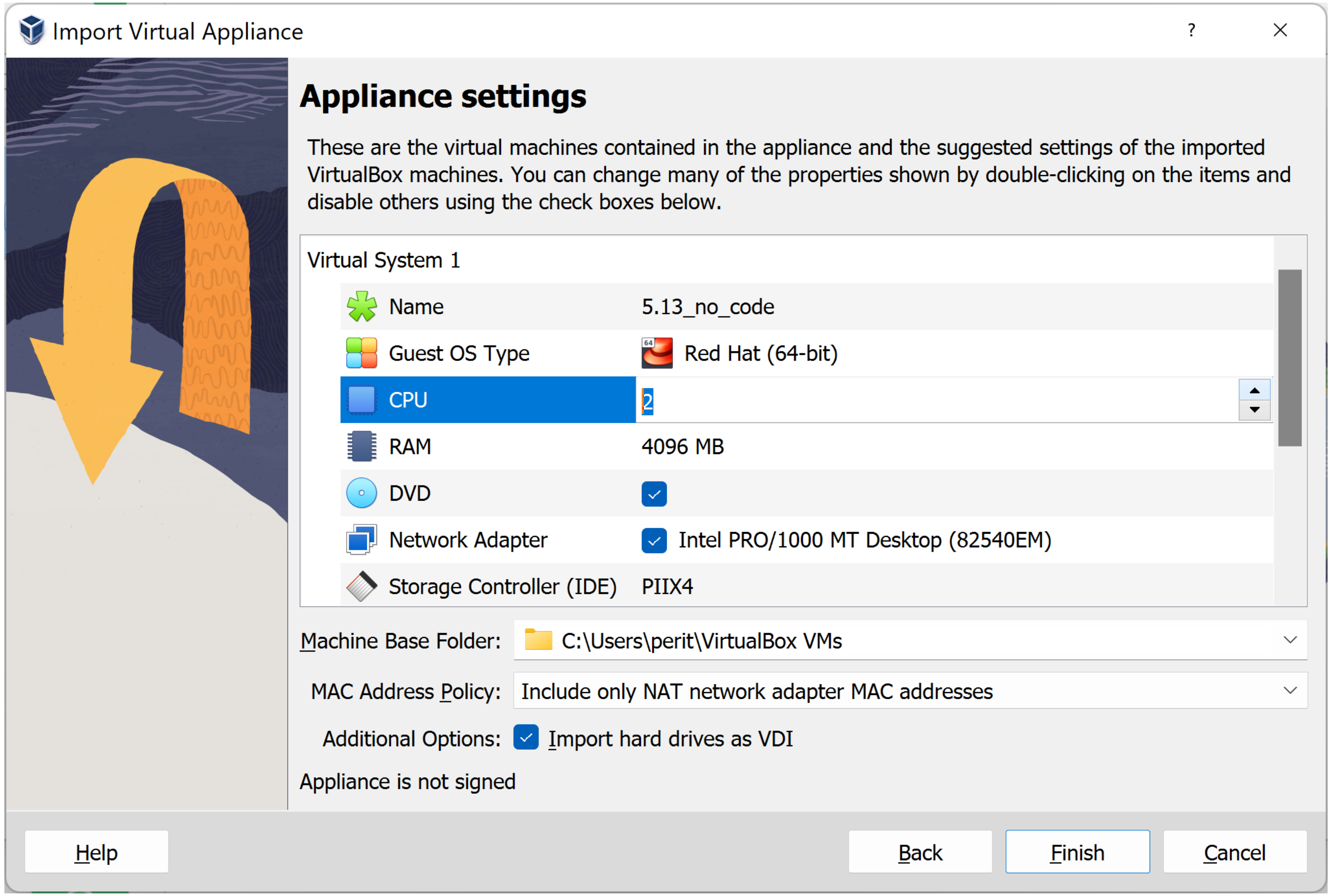


Review/Update VM Hardware Resource



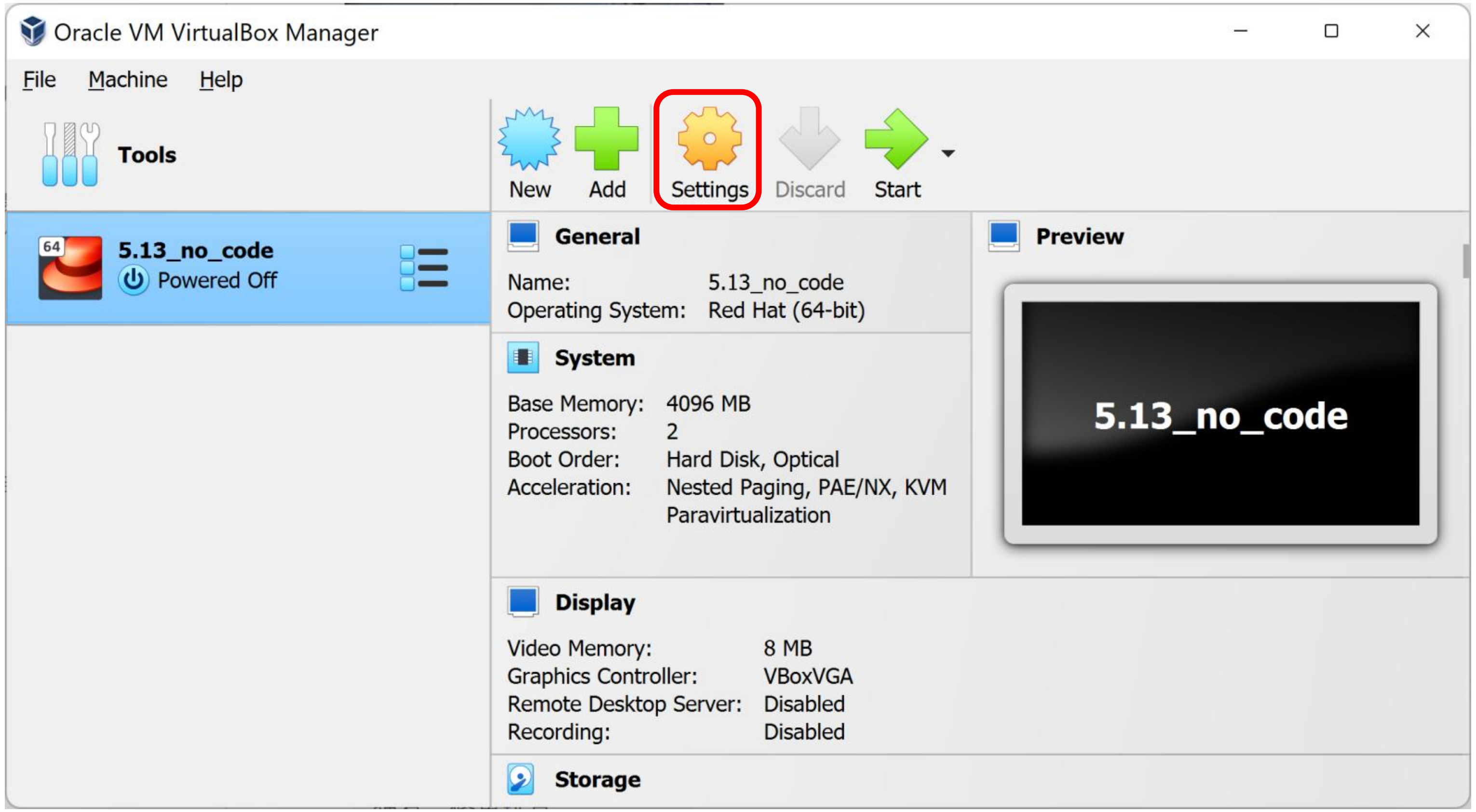
Review/Update VM Hardware Resource (2)

- You can optionally decide whether to increase the number of CPUs to 2 by double click on the corresponding row.
- If your physical machine is not very powerful, keep the virtual CPU count at 1.



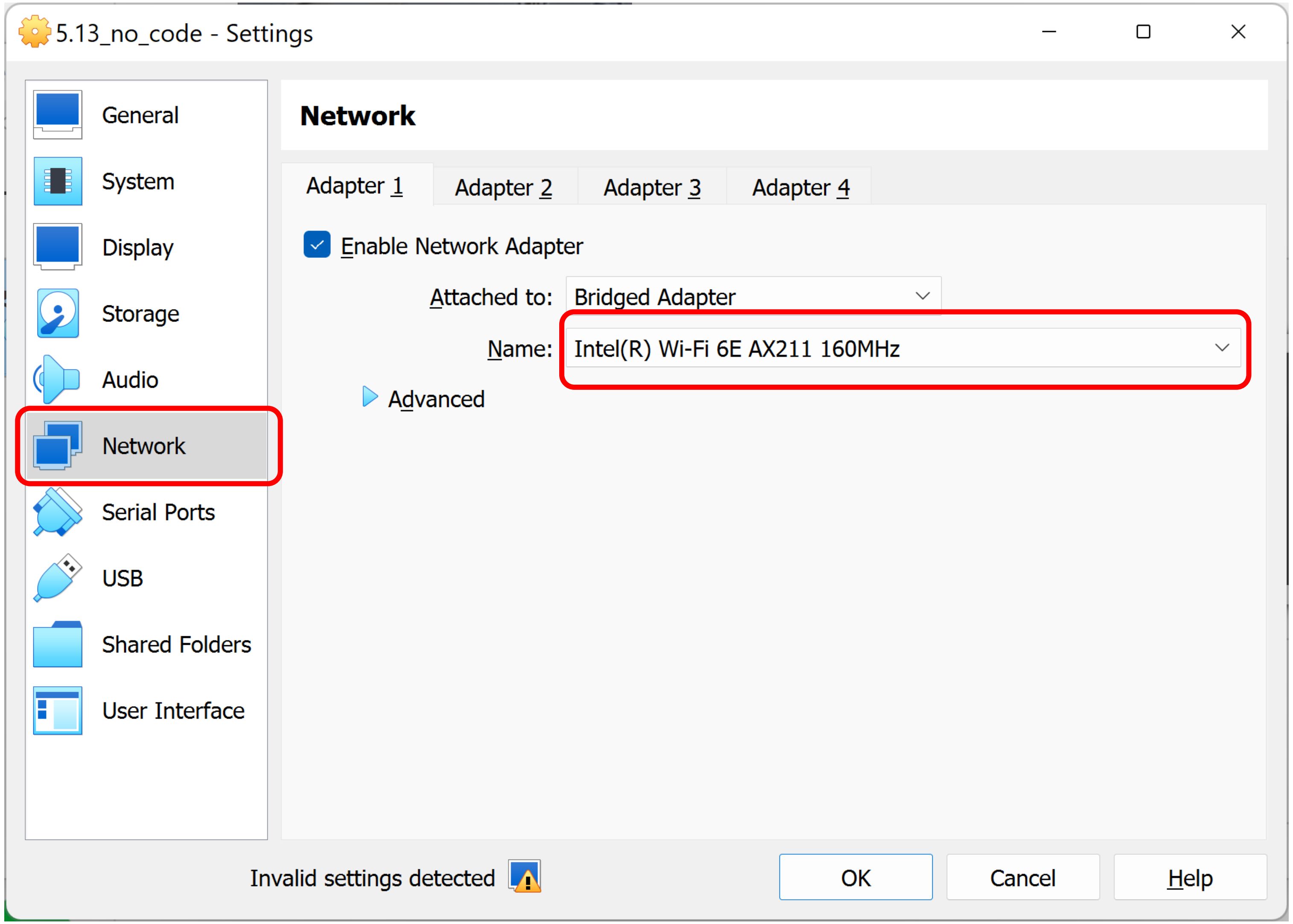
Update Network Setting

- The image was built on a machine which is different from yours.
- Click “Settings” to update the network configuration.



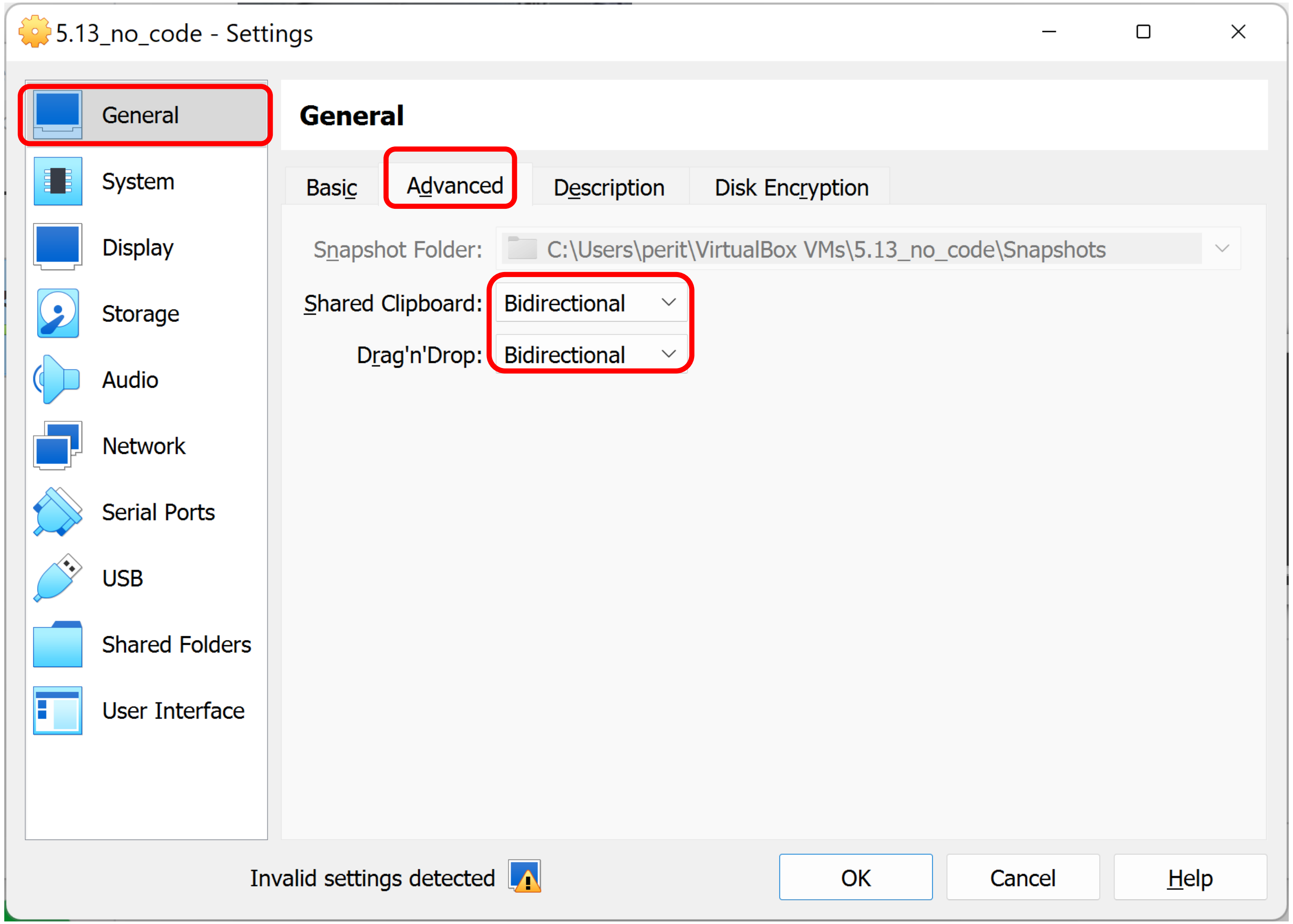
Update Network Setting

- The image was built on a machine which is different from yours.
- Click "Settings" to update the network configuration.
- Make sure the "Name" is selected to the correct network card.
- Click "OK" to save the change.

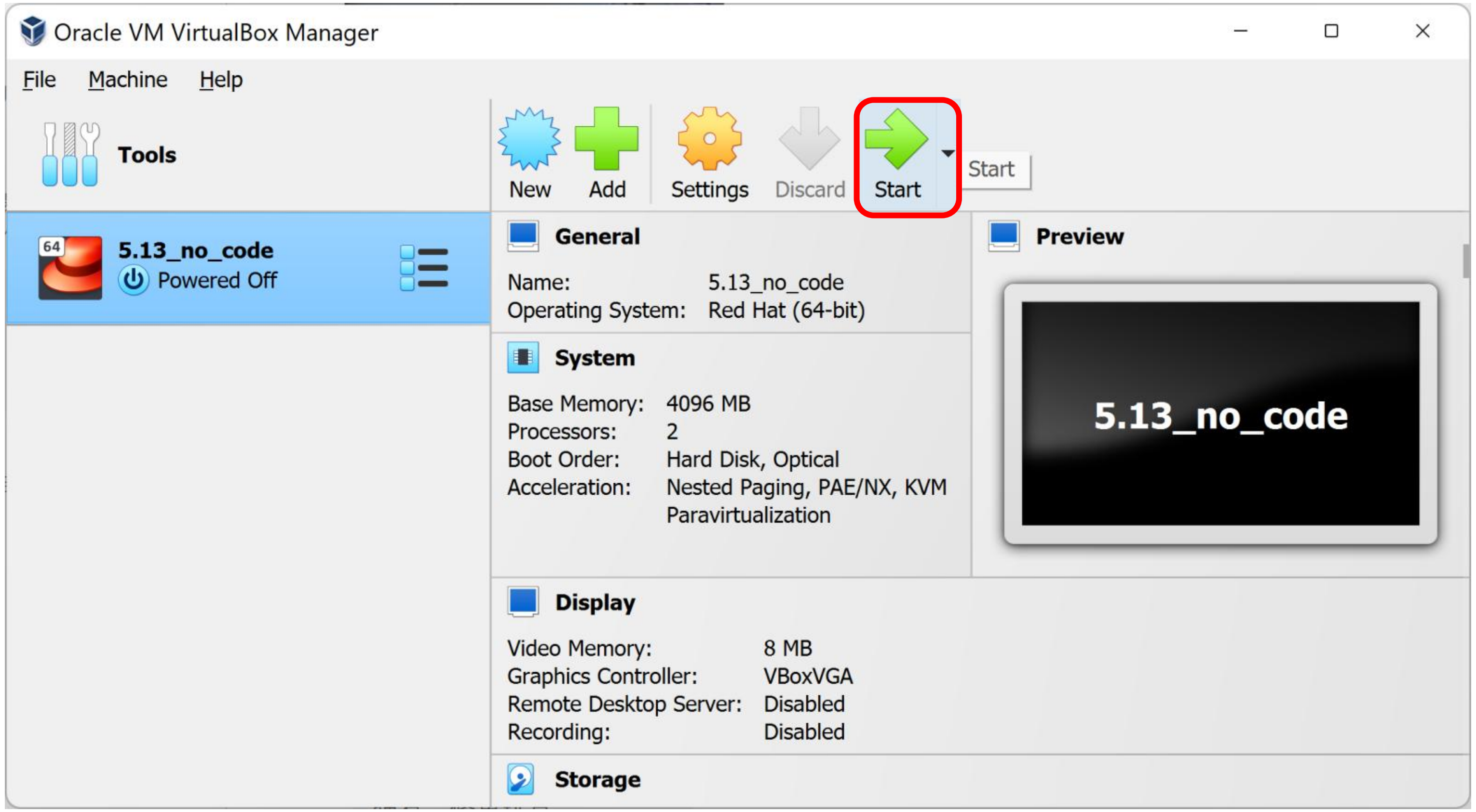


Update the Advanced Setting

- Enable the shared clipboard and drag'n'Drop helps you to work with the VM more conveniently.

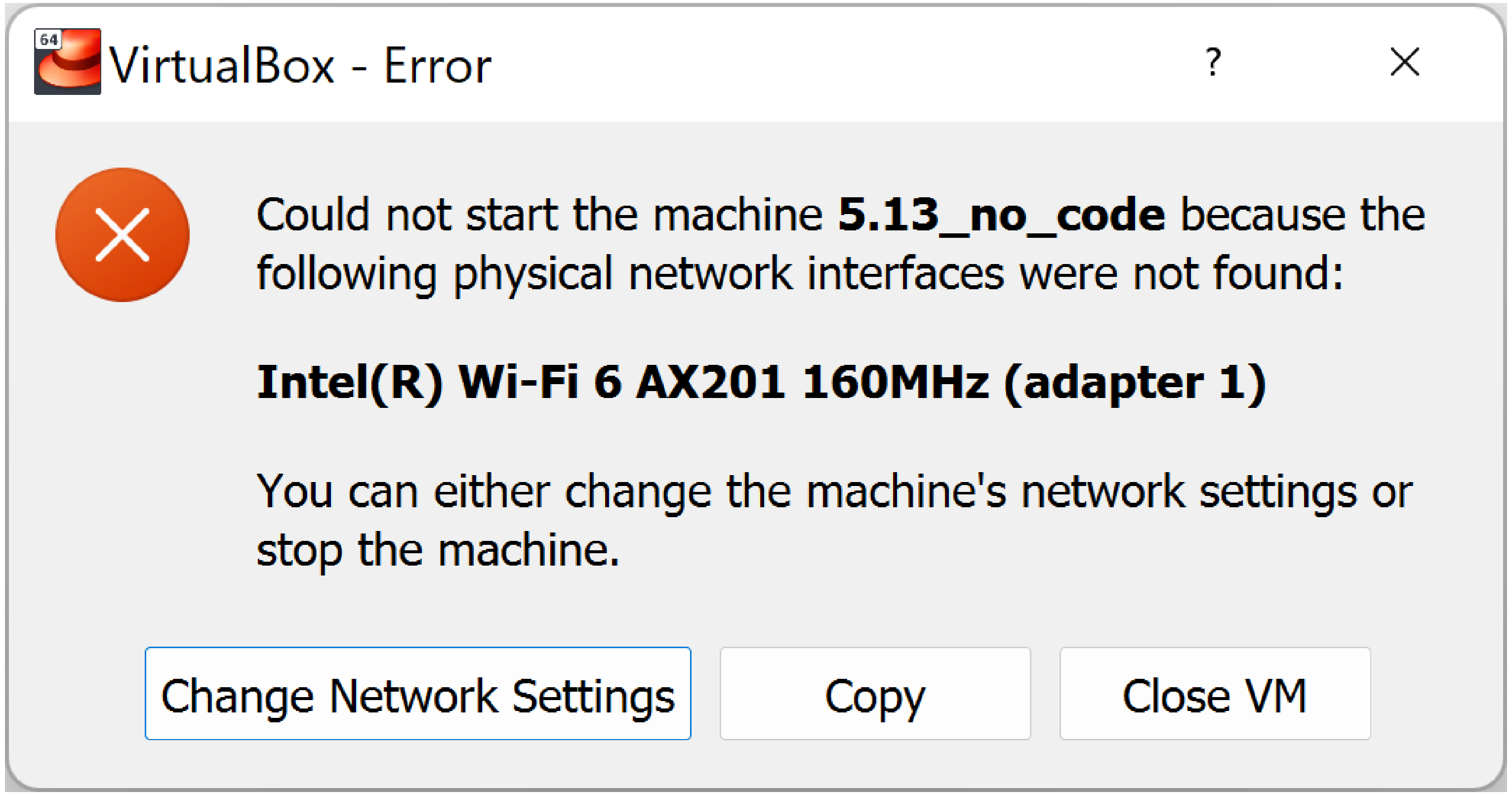


Start the Virtual Machine

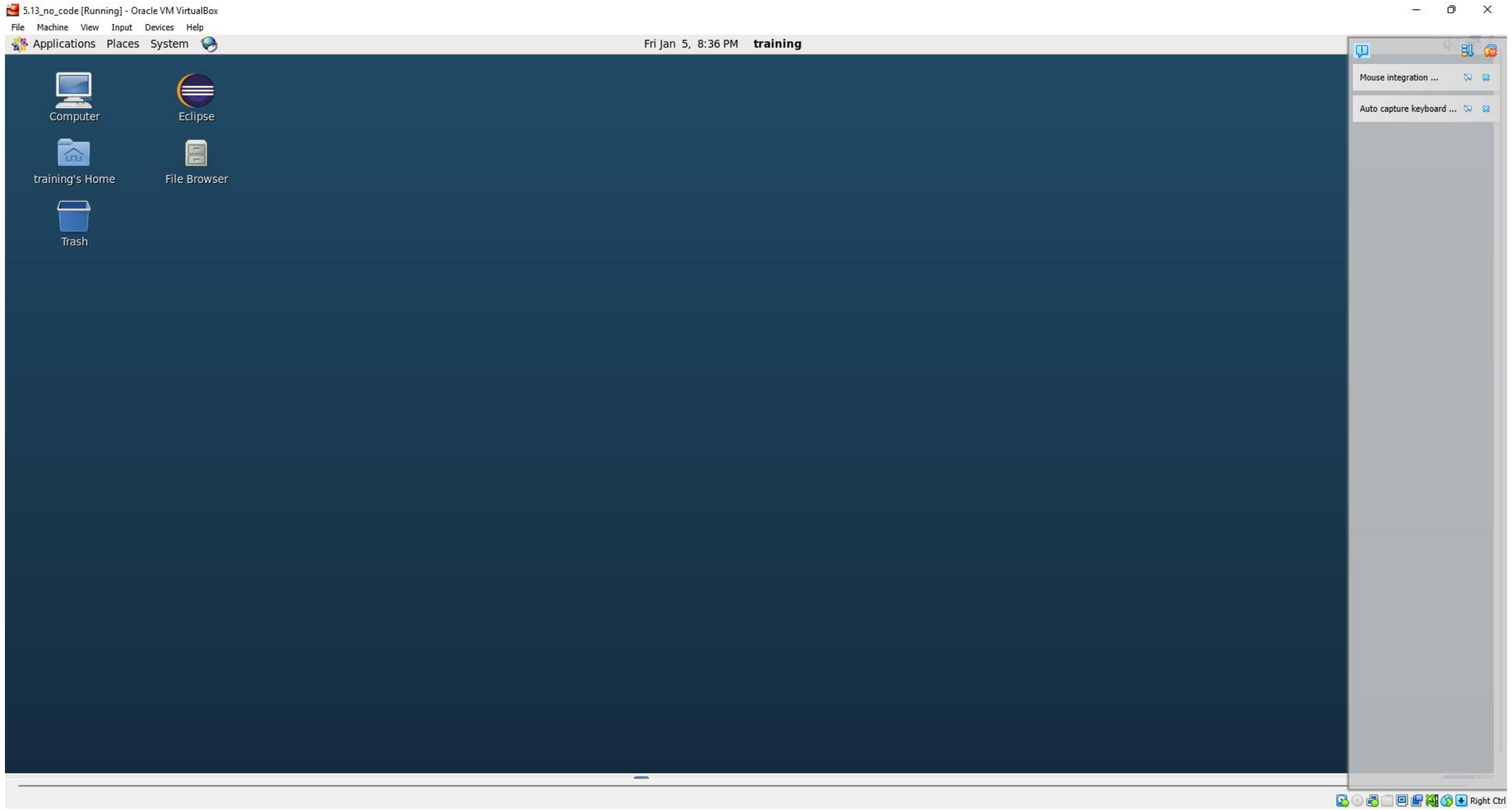


Encounter an Error?

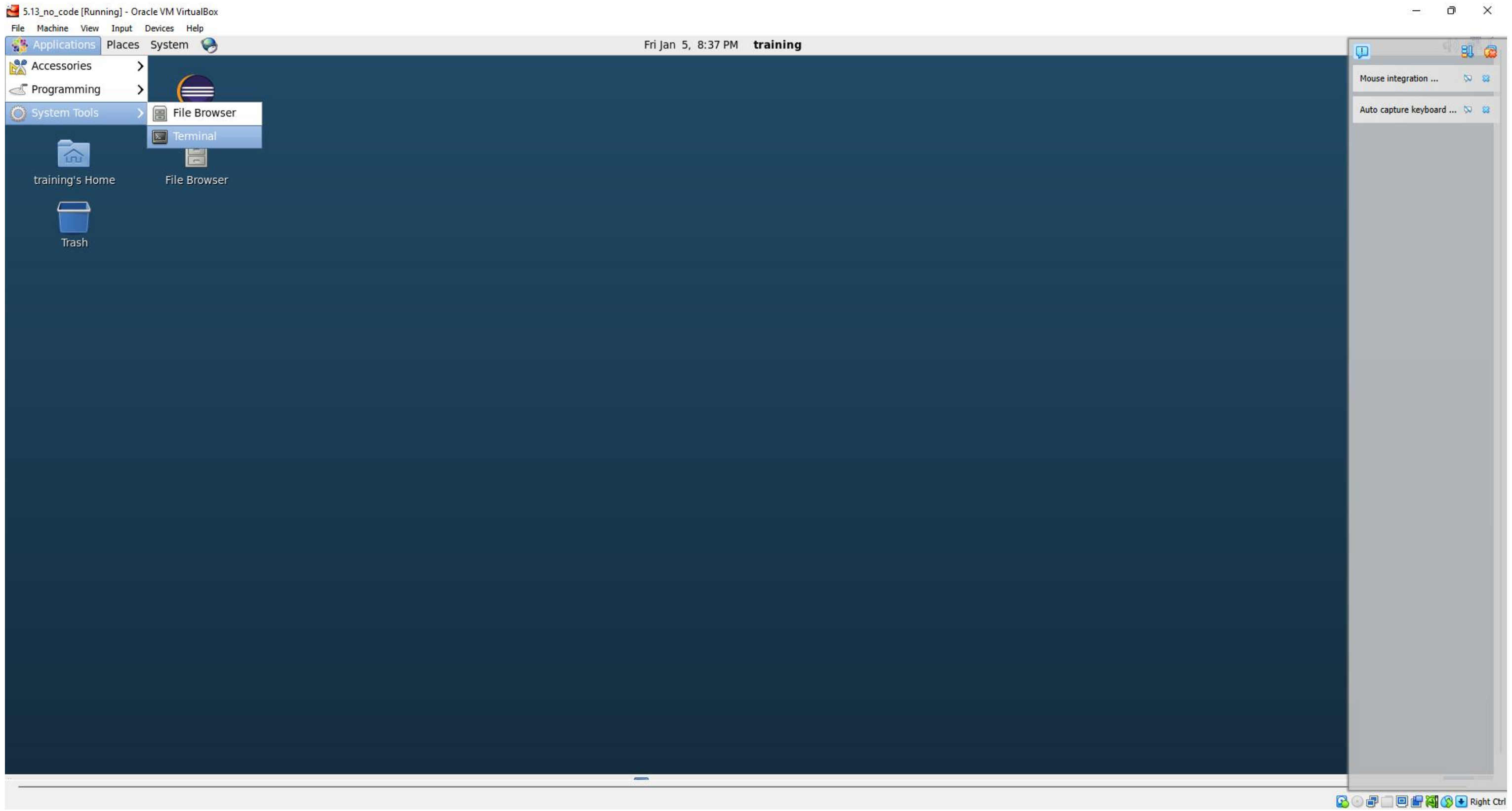
- It is normal if you encounter an error such as the physical network interface is not found.
- This image is built on a machine which is different from yours.
- The solution is to change the network settings.

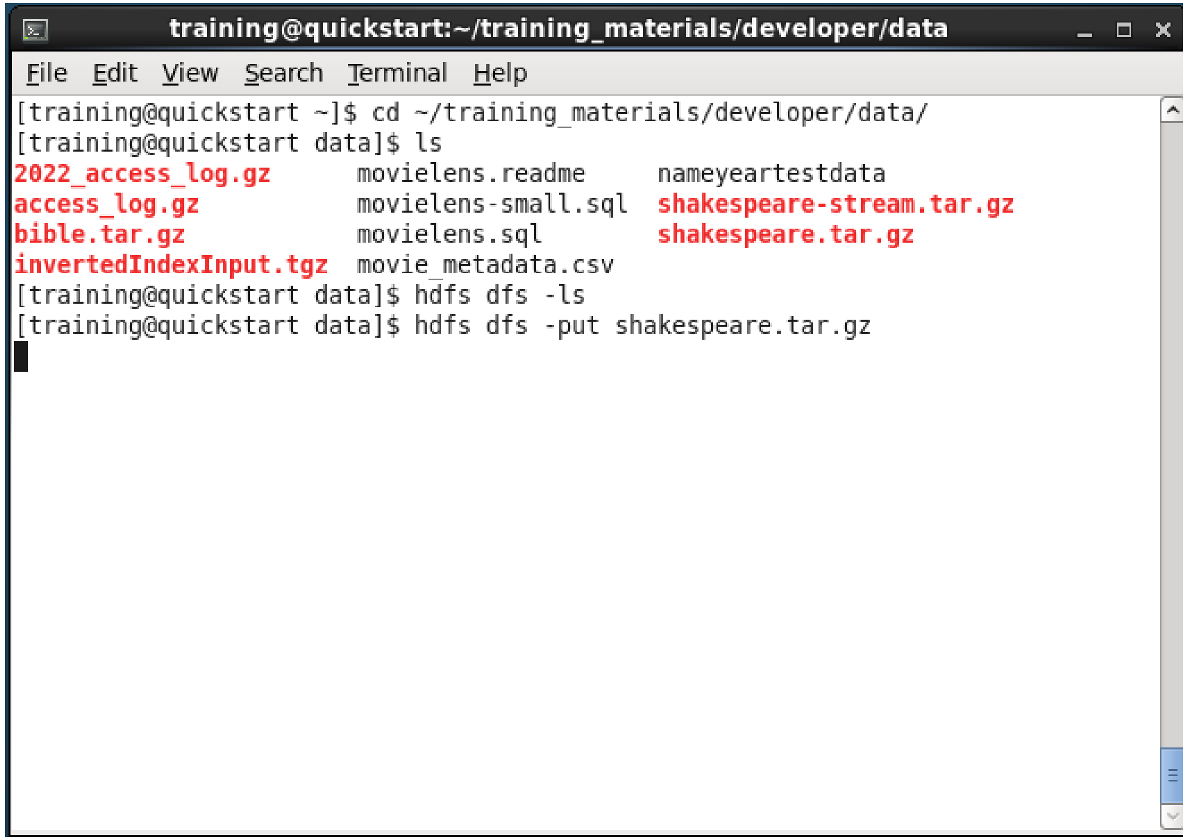


Virtual Machine Environment



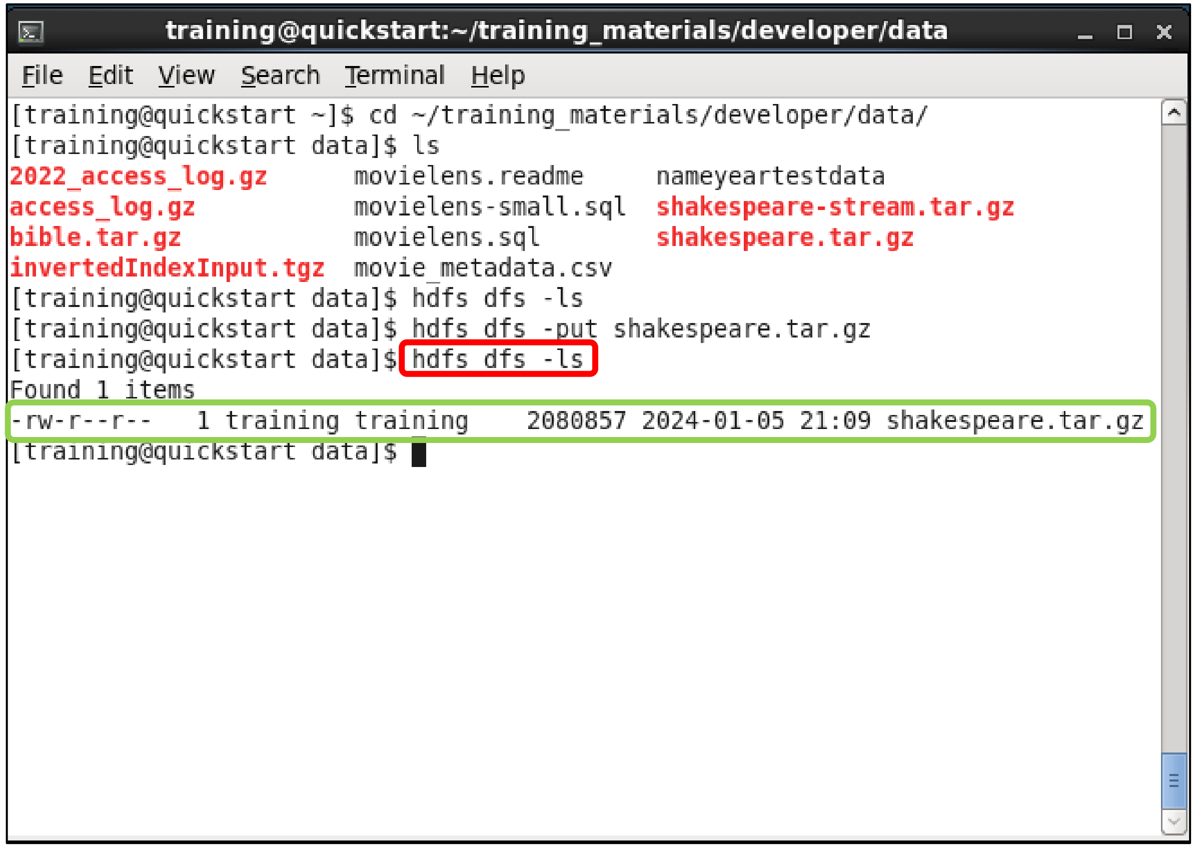
Open a Terminal





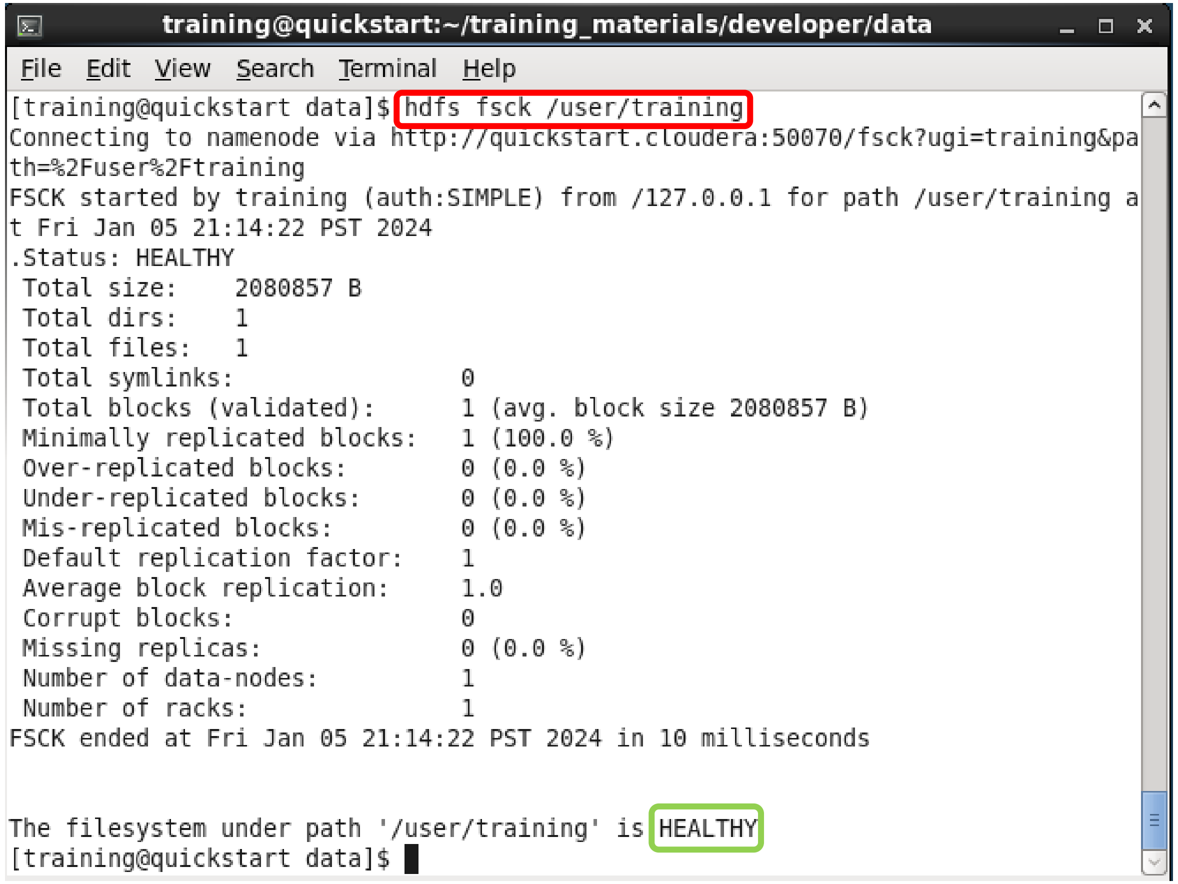
Upload File to HDFS

- Switch to the source folder and list the content:
  - cd ~/training_materials/developer/data/
  - ls
- Check what is in your Hadoop folder:
  - hdfs dfs –ls
  - (By default, it checks the user's home directory.) (In this case: /user/training)
  - Displays nothing when the folder is empty.
- Upload the file/folder:
  - hdfs dfs –put Shakespeare.tar.gz



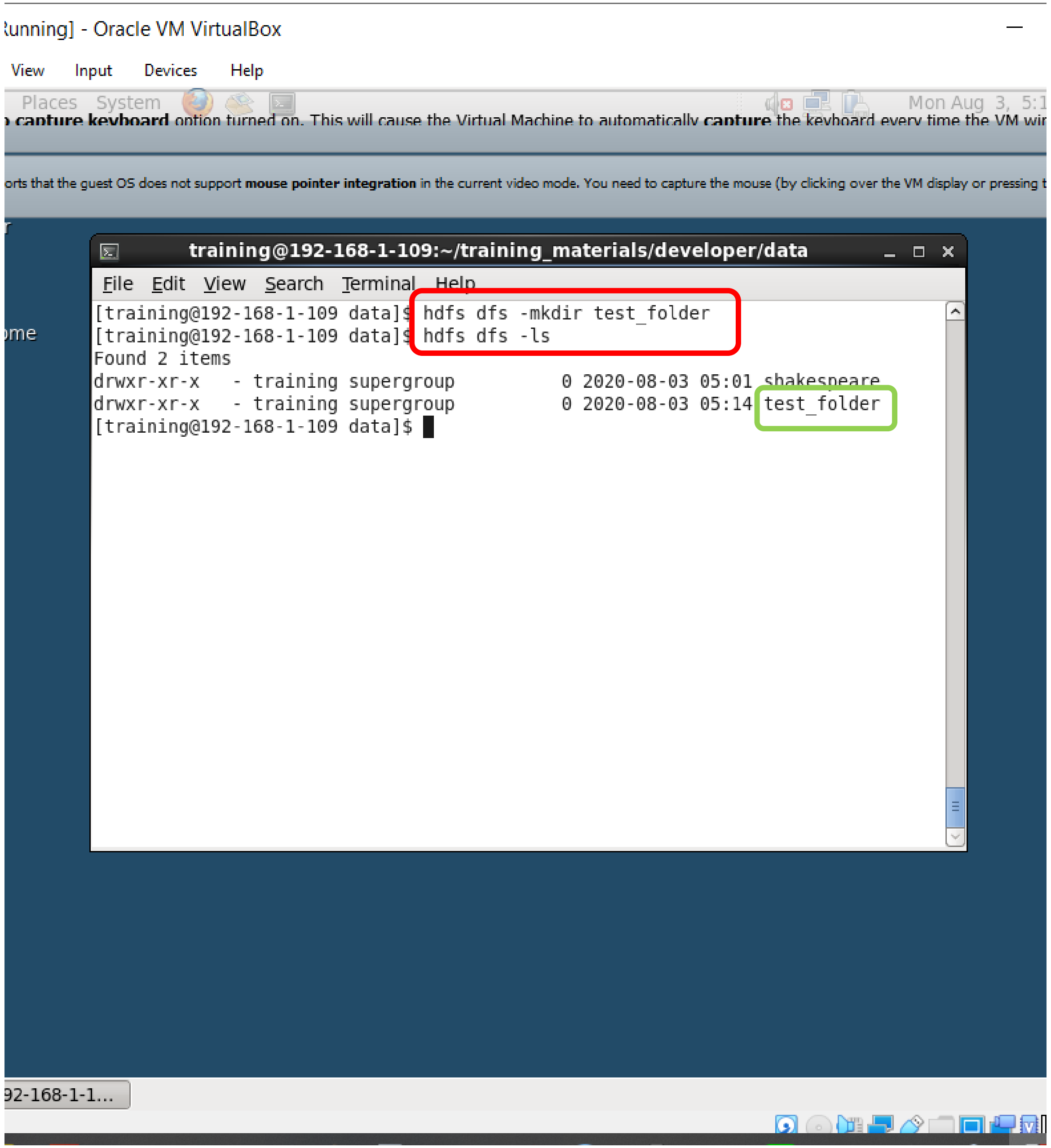
Check Hadoop Content

- Check what is in your Hadoop folder:
  - hdfs dfs –ls
  - You'll find the uploaded compressed file is there.



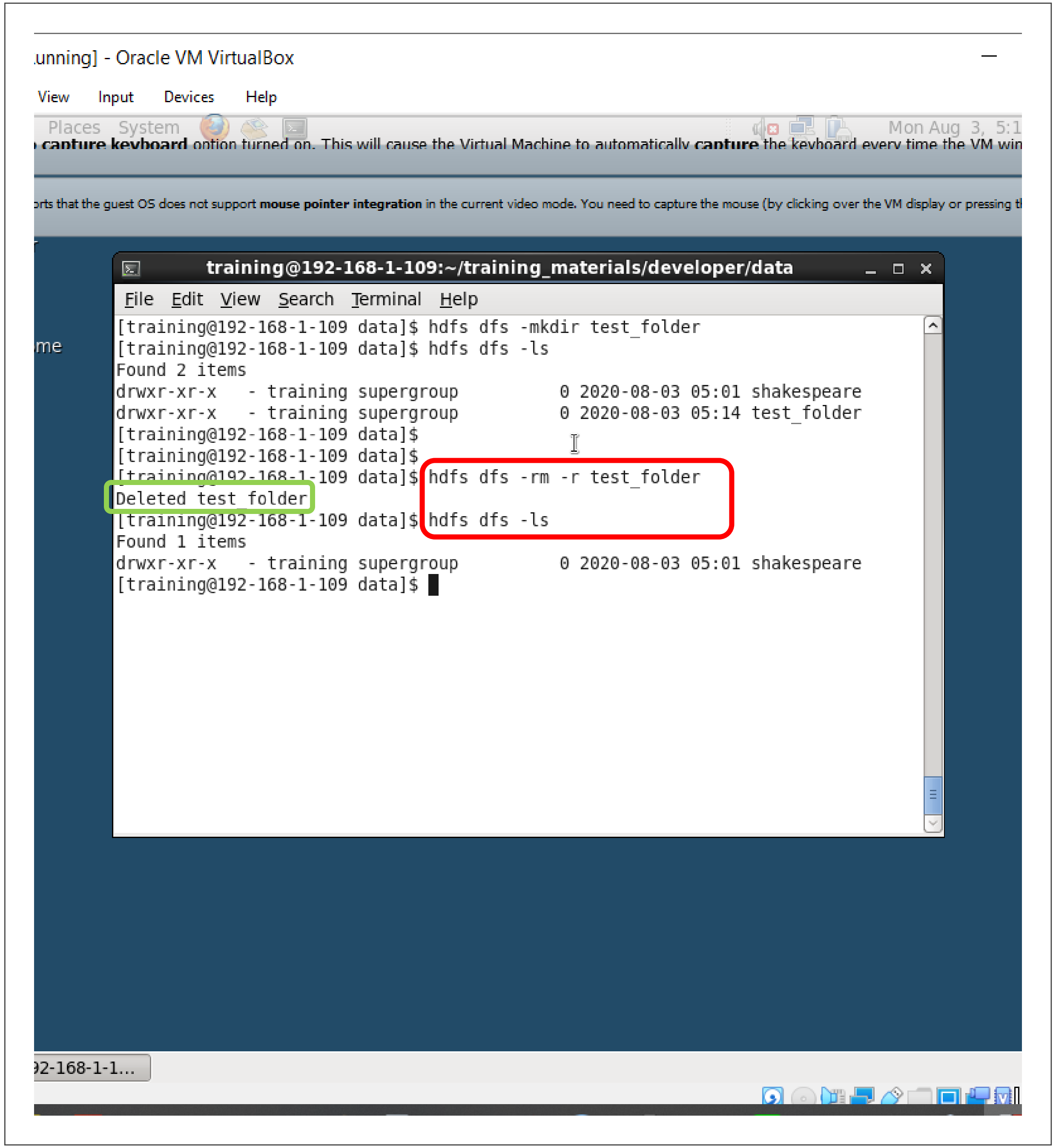
Check HDFS Healthy Status

- **hdfs fsck /user/training**



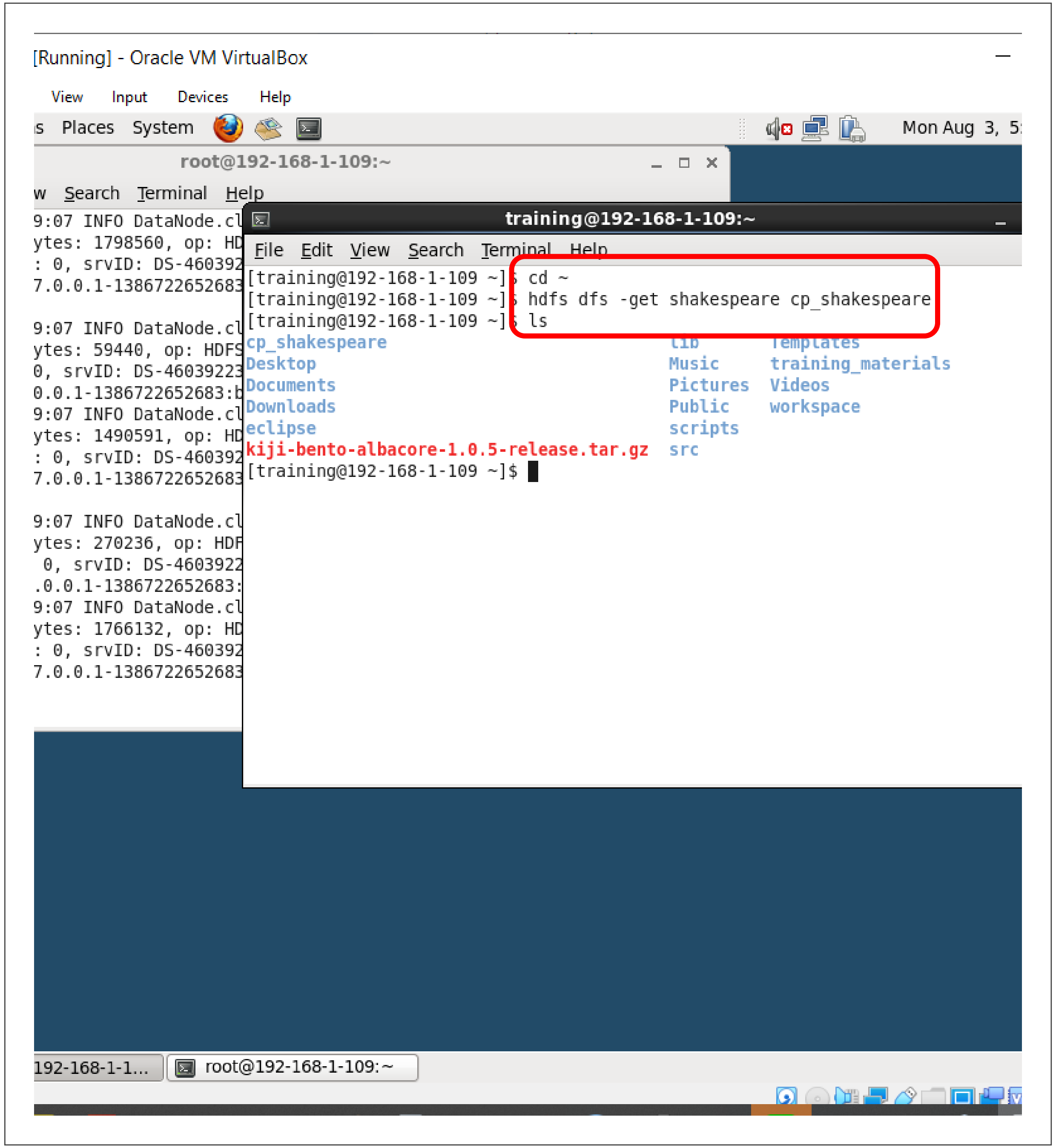
Create Folder

- **hdfs dfs –mkdir test_folder**
- **hdfs dfs -ls**



Remove Folder

- **hdfs dfs –rm -r test_folder**
- **hdfs dfs -ls**



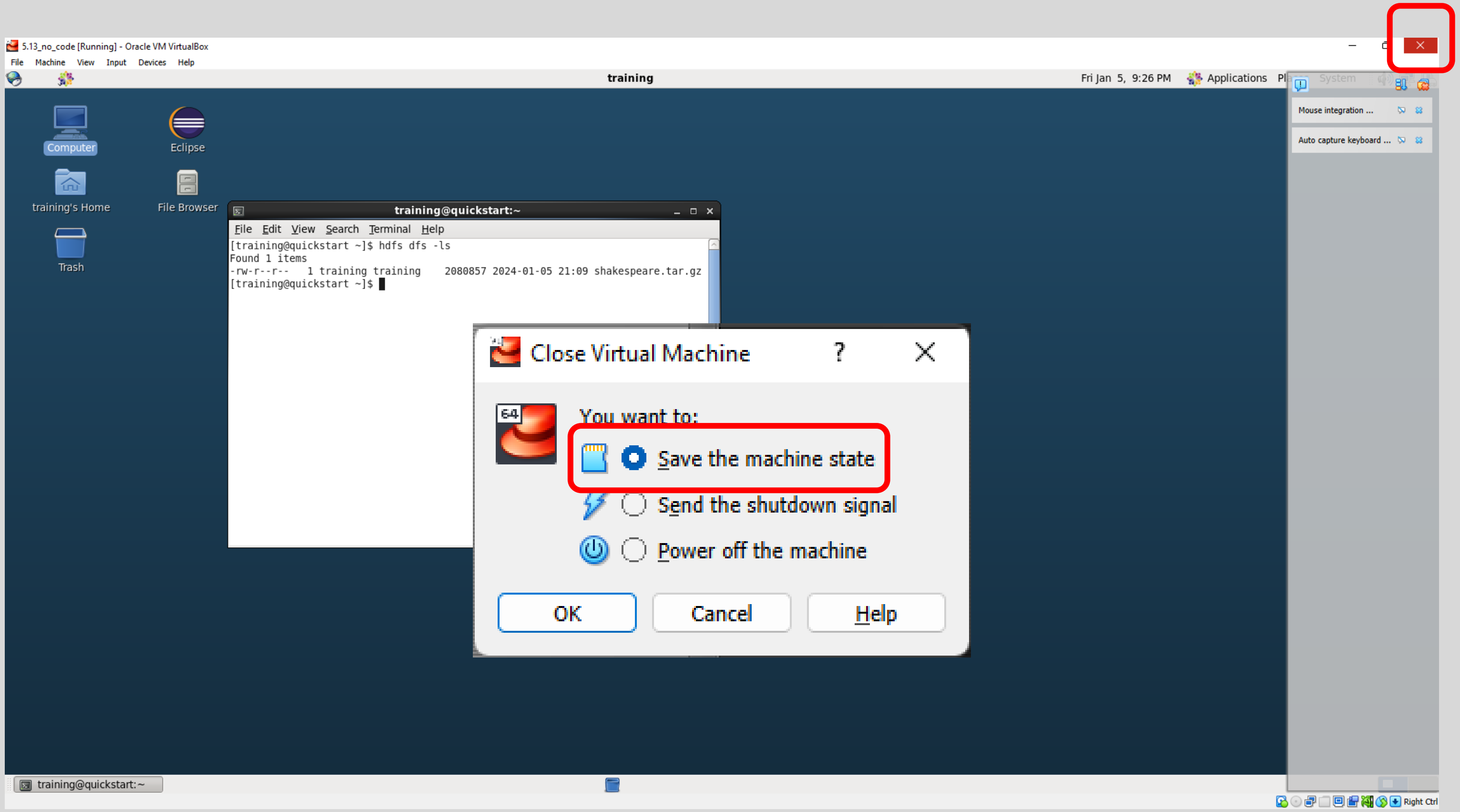
Download Files from Hadoop

- cd ~
- hdfs dfs –get shakespeare cp_shakespeare
- ls

Password for Entering the System

- If you got locked out by the screen saver, use the password “**training**” to get back in again.

Freeze the Machine State



- Freeze the state of the machine instead of powering it down.
- This will help you get everything loaded much faster the next time you enter the virtual machine.